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POSTER

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A Review of Text Accessibility Standards, Guidelines, and Font Tool Limitations

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Abstract

Typography remains an underexplored aspect of accessibility despite its significant impact on readability and inclusive design. While the Web Content Accessibility Guidelines (WCAG) address factors like color contrast, spacing, and scalability, it offers limited guidance on typographic variables such as font style, size, and letterforms. Through this study, we investigated the state of typographic accessibility. We summarized 24 actionable guidelines from 10 academic and industry sources, evaluating how these guidelines are supported across seven widely used font libraries. Our findings reveal that font libraries lack accessibility support, showing a significant gap between knowledge and implementation. As one of the few studies to systematically assess text accessibility resources and tool support, we provide early guidance on integrating accessibility feedback into design workflows. We call for broader industry recognition and future efforts to reform standardized text accessibility guidelines and integrate their evaluation in design tools.

CCS Concepts

• **Human-centered computing** → **Accessibility**.

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1 Introduction and Related Work

Section 508 [21] defines text accessibility as the ability of digital content to be perceivable and usable by individuals with disabilities. In this context, an accessible font refers to a typeface that promotes legibility and readability for a broad range of users, including those with visual impairments, such as low vision, or cognitive conditions like dyslexia. Typography, the art of arranging text [9, 16, 24], is recognized for its impact on accessibility and readability [24]. And yet, it remains a largely unaddressed domain in accessibility. The limited WCAG [26] standards, which focus merely on color contrast and scalability, can account for this.

Prior HCI research has demonstrated that typographic characteristics, such as font style [18], font size [10, 19, 22], x-height [10], letter spacing [22], and character width [2, 16, 22], play a crucial role in readability and user comfort. Findings reveal that sans-serif fonts tend to outperform serif fonts in terms of reading speed and error rates, and larger x-heights, width, and letter spacing improve legibility, especially at smaller font sizes. Despite this evidence, accessibility standards like the WCAG do not address these nuanced typographic factors. Through this study, we have observed that the current accessibility resources on text are fragmented across various blogs and design community channels, making their usability and accessibility a challenge.

We followed a two-step process to evaluate the current state of text accessibility and answer the **RQ: What resources are available to support designers in creating accessible text?** First, we summarized typography guidelines and recommendations from academic resources, websites, blogs, and other relevant sources into a consolidated set of 24 practical design recommendations. Second, we evaluated seven widely used font libraries to assess their support for these guidelines. We found that current guidelines, though scattered and unorganized, align with prior findings on accessible typography. However, they remain unaddressed in industry standards and tools. While font libraries often include robust typography information, none offer accessibility validation or typographic feedback. Building on prior work that advocates for improved in-built accessibility support [11] and findings from a preliminary study with UI/UX designers, we propose an initial design solution in the typography domain. Although it remains only a conceptual prototype, we encourage the industry to further explore this domain. This solution represents the first step in providing better support for typographic accessibility during the initial design stages and informing future directions for integrating tools and developing guidelines.

2 Guideline Summary

We conducted a series of searches using keywords like “accessible text”, “text accessibility guidelines”, and “readability” on Google (for blogs and websites) and the ACM library (for academic papers). While limited academic work has produced actionable guidelines, we found multiple Google sources, including WCAG 2.2 [26], Section 508 [21], WebAIM [25], the Readability Group [23], Scope [20], and designer-focused platforms such as Google Fonts [8] and Protypo [17]. We focused on extracting practical design guidelines and organized the results into six categories: font style, size, spacing, alignment, color contrast, and general readability. In our paper, we use the term “guideline” to refer to practical recommendations,

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including both formal standards (e.g., WCAG) and emerging best practices from grey literature. While not all are equally validated, they reflect what designers currently rely on. Table 1 consolidates all the guidelines collected, along with their source(s) and the category to which they belong.

2.1 Findings

Alignment guidance strongly favored left-aligned text and disapproved center and right alignments for long blocks of text, as they were difficult to track. *Color contrast* guidelines in all sources matched WCAG’s 4.5:1 ratio for standard text and 3:1 for large text (text size above 18pt or 14pt bold).

Font style was a popular topic of discussion in the sources. Many recommend using sans-serif fonts for body text, serif fonts only for headings, and avoiding decorative or script styles. Beyond font families, several sources emphasized native font characteristics that influence readability, including counters, letter ambiguity, mirroring, and x-height. Open counters (the internal white space within letters like ‘o’ or ‘e’) enhance legibility, while closed counters make letters appear dense and harder to read at small sizes. Letter ambiguity occurs when characters like lowercase ‘l’, uppercase ‘I’, and the number ‘1’ look nearly identical, potentially confusing users with low vision. Mirroring describes when letters are visual mirror images of one another, such as ‘b’ and ‘d’, which can confuse readers with dyslexia. X-height (the height of a lowercase ‘x’ from the baseline) affects how large and readable a font appears. Larger x-heights are generally known to improve legibility. However, some sources suggest using medium x-heights for body text and avoiding overly large ones that can make upper and lowercase letters indistinguishable. While these traits are widely acknowledged, no formal parameters currently define ideal values for them.

Some sources offered general *Readability* recommendations rather than font-specific advice, such as maintaining proper layout at 200% zoom (WCAG), keeping line lengths between 40–120 characters (WebAIM, APA), avoiding motion in text, and using correct quotation marks for screen reader compatibility. These guidelines aim to minimize cognitive load and maintain clarity across various viewing conditions. *Font size* was primarily discussed in relation to Section 508. They unanimously agreed that 16px (pixels) or 1rem (root em, a relative unit of measurement) is the ideal minimum value for body text. At the same time, some allowed smaller sizes, such as 12px to 13px, for short text (labels and captions). *Spacing* between letters, lines, paragraphs, and words was widely discussed in various resources, including the WCAG. These parameters have a significant impact on overall readability, particularly for older adults, individuals with dyslexia, and those with low vision. The sources recommend ideal metrics for each parameter: Letter spacing should be at least 0.12 times the font size (12%), line height at a minimum of 1.5 times the font size (150%), paragraph spacing should be at least double the font size (200%), while word spacing stands at 0.16 times the font size (16%).

2.2 Summary

In summary, WCAG appeared most frequently, being cited in seven guidelines, particularly those related to contrast and spacing. Meanwhile, nearly half of the recommendations focus on font readability,

including x-height, open counters, and character ambiguity. Table 1 summarizes 24 practical text accessibility guidelines from ten widely recognized references into a centralized resource, serving as a key contribution of this study.

3 Feature Analysis of Font Libraries

After collecting and synthesizing the text accessibility guidelines, the next milestone of the research study was to examine how these guidelines are incorporated into designers’ tools. A detailed feature analysis of websites containing font libraries was the ideal decision, as they are platforms dedicated to typography, provide extensive details on fonts and type, and contribute to designers’ decision-making processes. This rationale was based on personal experience, logical assumptions, and standard design practices observed. The primary goal of the feature analysis was to examine the level of typographic details the websites provide, and the extent of accessibility information associated with it.

We used search keywords such as “best font websites” and “famous font websites” on Google and selected the top seven organic results. The final chosen websites were: Google Fonts [8], daFont [3], Adobe Fonts [1], Font Squirrel [5], MyFonts [13], Fontspace [7], and Fontfabric [6].

3.1 Findings

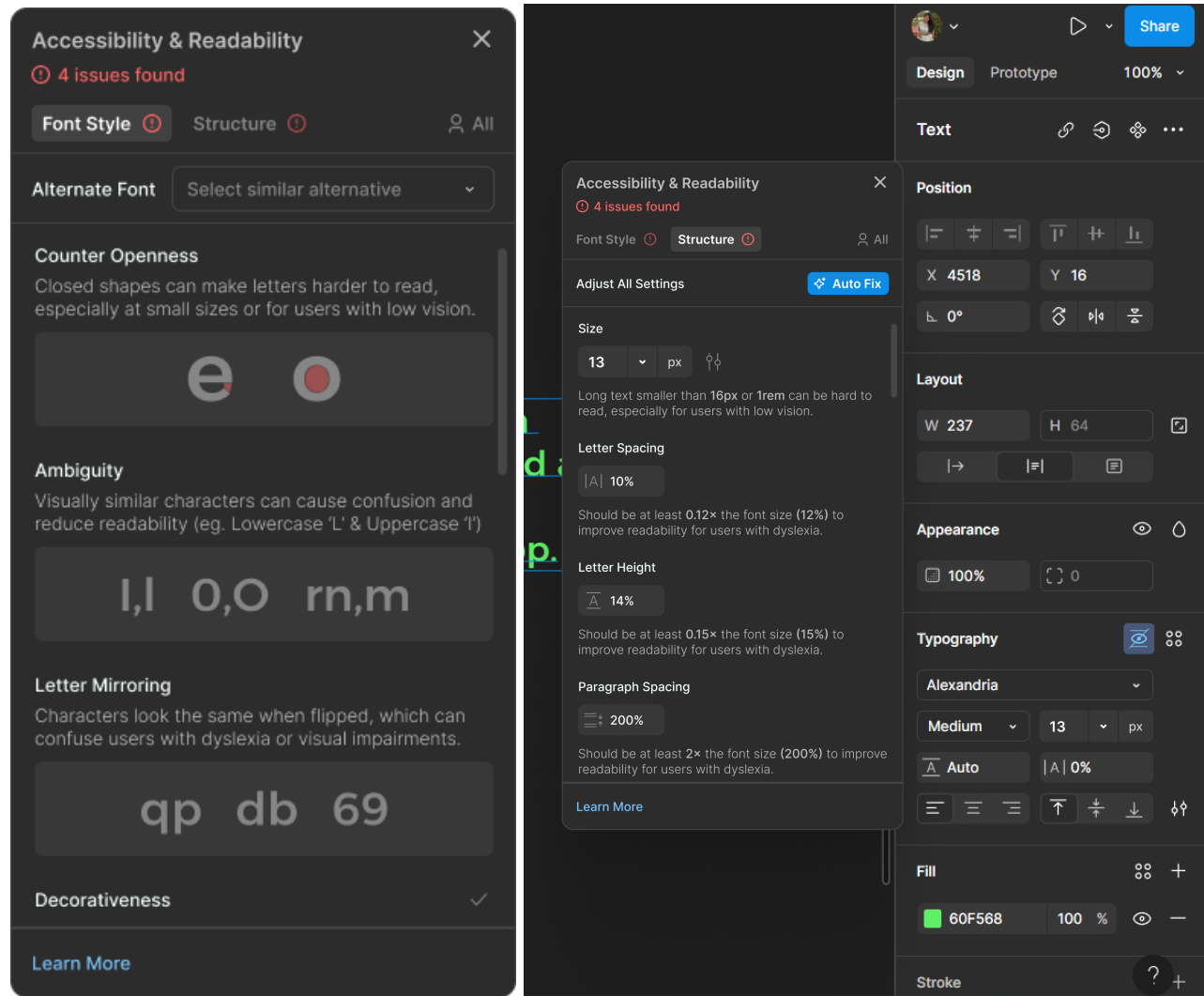
Table 2 presents a union of features identified across the seven websites and compares each website to this union. The features used for evaluation were: preview text based on adjustable characteristics (size, line height, letter spacing, background and text color, case, weight, width, and alignment), specimen previews, font metadata (style and designer), search and filter functionality, informational resources (particularly for accessibility), font pairing suggestions, and others listed in the table.

Of the seven websites, only three provided resources and educational content (Google Fonts, Adobe Fonts, and MyFonts). Google Fonts was the only one to offer extensive details on accessibility and readability. However, this information was hidden in the ‘Knowledge’ window, far from the font browsing section. The platform also allows users to review and adjust various characteristics under a ‘Type Tester’ section, but does not validate those adjustments for accessibility. The other two websites provided resources unrelated to accessibility. Some of the websites analyzed offer extensive features, including a community channel, font pairing suggestions, advanced filtering options, trending fonts, and one of them also features a live chat option. However, neither of them associated the fonts with accessibility.

This analysis highlights that while font library websites offer robust exploration features, none successfully guide users towards making the right font choices.

4 Preliminary Progress and Next Steps

To understand how designers currently work around text accessibility and navigate challenges, we conducted a preliminary informal study with practicing UI/UX professionals. We used a screening survey to collect data from 21 designers of different levels of experience, academic background, and geographic origins. We filtered survey



(a) Text Accessibility checker with font style tab selected

(b) Tool inside Figma's interface with font structure tab selected

Figure 1: Hi-fidelity dark mode prototype of text accessibility checker

responses based on designers' experience in working with text accessibility. Those who admitted having knowledge and hands-on experience with it (eight of 21 designers) were invited to participate in interviews. Our interview questions were used to explore their current text decision workflows, awareness and use of guidelines, pain points with existing font tools, and desired features for accessibility support.

Our interviews revealed that while designers prioritized accessibility, their text decisions were primarily guided by intuition and color contrast. Even those with strong typographic knowledge often lacked the necessary tools or support to apply it, frequently constrained by workplace limitations or limited awareness. Participants expressed interest in having more explicit guidance and support built directly into their design tools.

Based on these findings, we propose developing a dedicated typographic accessibility tool (see Figure 1) integrated within Figma or similar design platforms. Such a tool should assess the accessibility of text against the guidelines summarized in this study. It should provide feedback on font style and structure based on established guidelines, enabling designers to make informed decisions without leaving their workflow. It could also offer preset recommendations, warnings for potentially ambiguous characters, and corrective measures.

The proposed solution responds to the suggestions from previous work on built-in accessibility support [11]. Although it is currently a conceptual prototype, our next step is to validate its effectiveness. Testing it with primary users, who are UI/UX designers, would demonstrate the tool's real-world usability, adoption, and impact on design processes, facilitating further refinements. We offer this

work as a starting point and call on the industry to further explore and address the typographic accessibility gap.

5 Discussion

After reviewing current text accessibility guidelines, we noticed that they align closely with prior HCI research on readability and accessibility. Both confirm that various typographic factors, such as size, spacing, style, and color, influence accessibility. And yet, they remain underrepresented in official standards, such as WCAG, which primarily address text color contrast and responsiveness. This disconnect between research and regulation leaves a critical gap in how accessibility is understood and implemented by designers. Feature analysis of font libraries further amplifies this gap by highlighting the limitations of tools and resources in supporting accessible choices. While these platforms are designed to help users explore and select fonts, they provide minimal accessibility guidance, rarely referencing readability best practices, or offering validation based on accessibility criteria. This reflects a broader issue noted in previous work: accessibility considerations are often introduced late in the design process and depend on third-party tools [11].

Prior studies highlight Figma's reliance on third-party accessibility plugins, which may not be very efficient. While recent updates include built-in accessibility features for color contrast [4], typography remains largely unaddressed. This gap may stem from the perception that text accessibility is either too subjective or complex to automate. However, the guidelines gathered in this study show that many text properties, such as decorative-ness, x-height, and counters, can be evaluated. With advancements in technology and AI, systems can be developed to assess these attributes against recommended standards. Fontbase [12], for instance, already allows users to filter fonts based on properties like proportion, width, and x-height, demonstrating the feasibility of automated text evaluation. Such automation would reduce designers' reliance on subjective judgment and fragmented resources.

Together, these findings highlight the importance of expanding the scope of text accessibility standards and their implementation in design tools.

6 Conclusions and Recommendations

Through this study, we explored the field of accessible typography, summarizing empirical research and providing recommendations for best practices. Although typography directly affects readability, current tools offer little to no support for helping designers make informed choices. Given the limited research in this area, our work provides an early step toward addressing this gap. We want to encourage researchers to further explore this space. Lastly, integrating guideline-based typographic checks into design tools can move the industry toward more inclusive practices that better support a diverse range of users.

We further advocate for a reformed industry-wide approach to text accessibility, facilitated by better standardization, technological evaluation of the standards, and integration of this technology in primary tools. Large-scale user studies and real-world evidence should support the reformed guidelines. While this study focuses on commonly cited guidelines, many of them originate from Western

contexts and may not generalize to other writing systems. For example, left-aligned text is often recommended, but proper alignment is standard in languages like Arabic. Emerging work in HCI has begun to explore cultural differences in accessibility and the limitations with Western-developed accessible design resources [14]. A recent study highlighted designers from the Global South encountered challenges when exploring text design that aligns with their culture due to WCAG's limited text style recommendations (including for other languages) [15]. Future iterations of this research will aim to include region-specific design tools and guidelines to better account for linguistic diversity, and other considerations such as a broader range of user demographics, device constraints, and situational impairments.

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A Appendix

Table 1: Publicly Available Text Accessibility Guidelines

Category	Guideline Summary	Source(s)
Alignment	Left-aligned text is preferred.	Readability Group, WebAIM
Alignment	Avoid fully justified text due to inconsistent spacing.	Readability Group, WebAIM, APA
Alignment	Avoid long blocks of centered or right-aligned text.	WebAIM, APA
Color	Normal text should have a contrast ratio of at least 4.5:1.	WCAG
Color	Large/bold text should have at least 3:1 contrast ratio.	WCAG
Font Style	Use sans-serif fonts for digital content. Serif fonts may be used for headings.	Section 508, Scope, Readability Group, Fontbase, Google Fonts
Font Style	Avoid italic, oblique, decorative, and script fonts in body text.	Scope, Readability Group, Section 508
Font Style	Fonts with large x-heights improve legibility, especially at small sizes.	Readability Group, Fontbase, Prototyp
Font Style	Select x-height based on context: medium x-height for body copy; slightly higher for small text; avoid very high for paragraphs.	Insider Software
Font Style	Open counters improve character recognition.	Readability Group, Fontbase, Google Fonts
Font Style	Avoid ambiguous and mirrored/confusing glyphs (e.g., “rn” vs. “m”, “1” vs. “l”). Ensure distinct letter pairs like ‘qp,’ ‘db,’ ‘0O,’ ‘il1l,’ ‘69,’ ‘a8,’ ‘a6,’ ‘6g,’ ‘rn, m.’	Readability Group, Fontbase, Google Fonts
Readability	Allow users to resize text up to 200% without breaking layout or losing content.	WCAG, Section 508
Readability	Line length should be between 40–70 characters per line (minimum 30, max 70).	APA
Readability	Fewer than 50 or more than 120 characters per line reduces readability.	WebAIM
Readability	Avoid blinking, animated, or scrolling text.	Readability Group, WebAIM
Readability	Underlines should only be used for hyperlinks.	WebAIM
Readability	Use proper quotation marks (not inch or foot marks) for assistive tech support.	APA
Size	Minimum font size should be 16px (1rem) for body text.	Scope, Readability Group, Section 508, Google Fonts
Size	Digital characters should be at least 3/16 inch tall (around 16pt).	Section 508
Size	Font size may vary by context—smaller sizes (e.g., 12–13px) are acceptable for captions or labels if readability is preserved.	Insider Software
Spacing	Line height should be at least 1.5× font size.	WCAG, APA, WebAIM
Spacing	Paragraph spacing should be 2× the font size.	APA, WebAIM, WCAG
Spacing	Word spacing should be at least 0.16× font size.	WCAG
Spacing	Letter spacing should be at least 0.12× font size.	WCAG

Table 2: Feature Comparison of Popular Font Libraries

Feature	Adobe Fonts	DaFont	Fontfabric	Font Squirrel	Fontspace	Google Fonts	MyFonts
Adjustable text properties (size, spacing, etc.)	Limited	Limited	Limited	No	No	Extensive	Limited
Advanced filters (x-height, properties, etc.)	Yes	No	No	No	No	Yes	Yes
Browsing categories (e.g., trending, seasonal)	Yes	No	Yes	Yes	Yes	No	Yes
Community support or help content	No	No	No	Yes	Yes	No	Yes
Custom text preview	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Educational resources or articles	Limited	No	No	No	No	Yes	Limited
Font metadata (style, designer)	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Font pairing suggestions	Yes	No	Limited	No	No	Yes	Yes
Font search and filters	Yes	Yes	No	Yes	Yes	Yes	Yes
Font specimen or visual design sample	Yes	Limited	Yes	Limited	Yes	Yes	Yes
Font usage instructions (embed, CSS, etc.)	No	No	No	No	No	Yes	Yes
“Fonts similar to” feature	Yes	No	No	No	No	Yes	Yes
Search by image (visual match)	No	No	No	Yes	No	No	No
Sorting options (e.g., popularity, release)	Yes	No	No	No	No	Yes	Yes
Comments	Resources available but not for accessibility	No accessibility, adjustable properties limited to size	No accessibility, adjustable characteristics limited to size, spacing, and height	No accessibility, longer navigation paths	Resources available but not for accessibility, unlabeled adjustable properties	Accessibility resources are available separately	Resources available but not for accessibility, unlabeled adjustable properties